Problem · Unit 2 · 10 points · 60 min · Standard exam level

P13 — Capsule descent under quadratic drag with two parachute stages: terminal speed, quadratures and the work of drag

*In a drop test, a capsule of mass m is released **from rest** from a hovering helicopter at the point O and falls vertically. Its drogue parachute is open from the release: the air exerts on the capsule the drag $-c_1 v^2$ along the motion, where $v = \dot{y} \geq 0$ is the speed and y the distance fallen, measured **downward** from O. When the speed reaches v_1, the main parachute opens instantaneously and the drag becomes $-c_2 v^2$, with $c_2 \gg c_1$. Gravity g is uniform and points down; buoyancy and the variation of air density with height are neglected.*

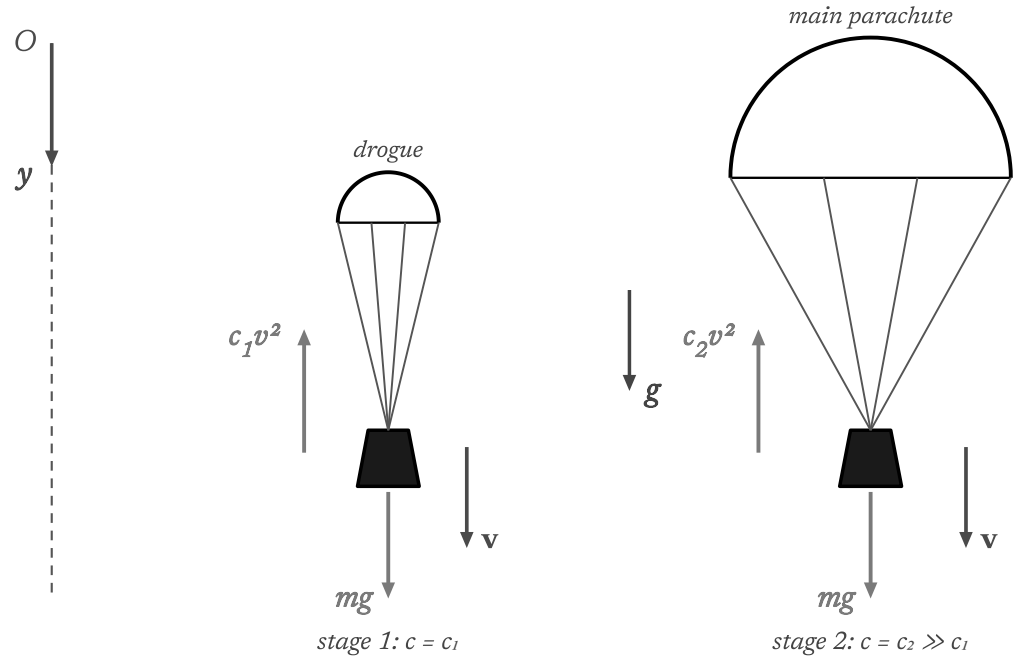


Figure 1. Two-stage descent: coordinate y downward from the release point O, velocity $v = \dot{y}$; forces on the capsule: weight mg (down) and drag cv^2 (up), with $c = c_1$ under the drogue (stage 1) and $c = c_2 \gg c_1$ under the main parachute (stage 2).

Setup

- Data: $m = 1000$ kg, $g = 9.81$ m s^{-2}. The parachutes are given by their terminal speeds: $v_{t1} = 50$ m s^{-1} (drogue) and $v_{t2} = 8$ m s^{-1} (main). The main parachute opens at $v_1 = 40$ m s^{-1}.
- Hypotheses: point particle, vertical motion (released with no horizontal velocity, no wind), constant drag coefficients within each stage, instantaneous opening.
- The time of stage 2 is counted from the opening, and Δy is the distance fallen since the opening.

Part (a) · 1 point

Degrees of freedom, forces and equation of motion

How many degrees of freedom has the capsule? Write the forces and the equation of motion for v.

Step 1. Count

With no horizontal force and no horizontal initial velocity the motion is along the vertical: 1 degree of freedom, the coordinate y, with unit vector $\mathbf{e}_y$ pointing **down**.

Step 2. Forces and Newton

Weight $mg\,\mathbf{e}_y$ (positive: it points down). Drag $-c|\mathbf{v}|\mathbf{v} = -cv^2\,\mathbf{e}_y$ for $v \geq 0$ (upward). Newton (Notes §4.3), with $\mathbf{a} = \dot{v}\,\mathbf{e}_y$:

$$m\dot{v} = mg - cv^2$$

KEY RESULTS

- 1 degree of freedom, $y(t)$, positive downward.
- $m\dot{v} = mg - cv^2$ (for $v \geq 0$).

Part (b) · 2 points

Terminal speed and $v(t)$ in stage 1

Define the terminal speed v_t and integrate the equation for the drogue stage, from rest at $t = 0$, to get $v(t)$ and $y(t)$.

Step 1. Terminal speed

The acceleration vanishes when $mg = cv^2$: $v_t = \sqrt{mg/c}$. With it the equation reads $\dot{v} = g\big(1 - v^2/v_t^2\big)$. For the drogue, $c_1 = mg/v_{t1}^2 = 9810/2500 = 3.924\ \text{kg m}^{-1}$.

Step 2. Separation of variables

For $0 \leq v < v_t$:

$$\frac{dv}{1 - v^2/v_t^2} = g\,dt$$

The antiderivative of the left side is $F(v) = v_t\,\operatorname{artanh}(v/v_t)$. Check with the chain rule: $\dfrac{dF}{dv} = v_t \cdot \dfrac{1}{1-(v/v_t)^2}\cdot\dfrac{1}{v_t} = \dfrac{1}{1 - v^2/v_t^2}$.

Step 3. Limits (Barrow's rule)

Integrating from $(t=0, v=0)$ to (t, v):

$$\left[v_t \operatorname{artanh}\frac{v'}{v_t}\right]_0^{v} = \left[g\,t'\right]_0^{t} \implies v_t \operatorname{artanh}\frac{v}{v_t} - 0 = gt \implies v = v_t \tanh\frac{gt}{v_t}$$

Since $\tanh < 1$, v approaches v_t but never reaches it in finite time.

Step 4. Position

$y = \int_0^t v\,dt'$. With $w = \cosh(gt'/v_t)$, $dw = \dfrac{g}{v_t}\sinh(gt'/v_t)\,dt'$, so $\tanh(gt'/v_t)\,dt' = \dfrac{v_t}{g}\dfrac{dw}{w}$ and

$$y = v_t \cdot \frac{v_t}{g}\left[\ln w\right]_{w=1}^{w=\cosh(gt/v_t)} = \frac{v_t^2}{g}\ln\cosh\frac{gt}{v_t}$$

KEY RESULTS

- $v_t = \sqrt{mg/c}$, i.e. $c = mg/v_t^2$; $c_1 = 3.924\ \mathrm{kg\ m^{-1}}$.
- $\displaystyle\int \frac{dv}{1 - v^2/v_t^2} = v_t \operatorname{artanh}\frac{v}{v_t}$ for $v < v_t$.
- $v = v_t \tanh\dfrac{gt}{v_t}$, $y = \dfrac{v_t^2}{g}\ln\cosh\dfrac{gt}{v_t}$.

Part (c) · 1.5 points

Speed as a function of the distance fallen

Obtain $v(y)$ for stage 1 directly from the equation of motion, without using t.

Step 1. Chain rule

With $v = v(y(t))$: $\dot v = \dfrac{dv}{dy}\dot y = v\dfrac{dv}{dy}$, so $v\dfrac{dv}{dy} = g\left(1 - v^2/v_t^2\right)$ and, separating, $\dfrac{v\,dv}{1 - v^2/v_t^2} = g\,dy$.

Step 2. Substitution

$w = 1 - v^2/v_t^2$, $dw = -\dfrac{2v}{v_t^2}dv$, so $v\,dv = -\dfrac{v_t^2}{2}dw$. Limits: $v = 0 \mapsto w = 1$, $v \mapsto w = 1 - v^2/v_t^2$:

$$-\frac{v_t^2}{2}\int_1^{1-v^2/v_t^2}\frac{dw}{w} = -\frac{v_t^2}{2}\Big[\ln w\Big]_1^{1-v^2/v_t^2} = -\frac{v_t^2}{2}\ln\left(1 - \frac{v^2}{v_t^2}\right) = g\,y$$

Step 3. Result

$$v^2 = v_t^2\left(1 - e^{-2gy/v_t^2}\right)$$

Cross-check: inserting $y(t)$ of (b), $e^{-2gy/v_t^2} = \cosh^{-2}(gt/v_t)$ and $1 - \cosh^{-2} = \tanh^2$, which is $v(t)^2/v_t^2$ of (b).

KEY RESULTS

- $v\dfrac{dv}{dy} = g\left(1 - \dfrac{v^2}{v_t^2}\right)$.
- $v^2 = v_t^2\left(1 - e^{-2gy/v_t^2}\right)$.

Part (d) · 1.5 points

Numbers for the drogue stage

If the main parachute never opened, how long and how far would the capsule fall before reaching $95\ \%$ *of* v_{t1}*? When and where does it actually reach* $v_1 = 40\ m\ s^{-1}$*?*

Step 1. $95\ \%$ **of the terminal speed**

$v_{t1}/g = 50/9.81 = 5.097$ s and $\operatorname{artanh} 0.95 = \tfrac{1}{2}\ln(1.95/0.05) = 1.8318$, so $t_{95} = 9.336$ s. From (c), $y = -\dfrac{v_t^2}{2g}\ln\left(1 - v^2/v_t^2\right)$ with $v_t^2/2g = 127.4$ m and $\ln(1/0.0975) = 2.3279$: $y_{95} = 296.6$ m.

Step 2. Opening of the main parachute

$\operatorname{artanh} 0.8 = \tfrac{1}{2}\ln 9 = 1.0986$, so $t_1 = 5.097 \cdot 1.0986 = 5.600$ s, and $y_1 = 127.4 \cdot \ln(1/0.36) = 127.4 \cdot 1.0217 = 130.2$ m. A numerical integration of the equation of motion gives the same values.

KEY RESULTS

- $t_{95} = \dfrac{v_{t1}}{g}\operatorname{artanh} 0.95 = 9.336$ s; $y_{95} = \dfrac{v_{t1}^2}{2g}\ln\dfrac{1}{1 - 0.95^2} = 296.6$ m.
- $v_1 = 0.8\,v_{t1}$ is reached at $t_1 = 5.600$ s, $y_1 = 130.2$ m.

Part (e) · 2 points

Stage 2: the main parachute

At the opening $v_1 > v_{t2}$. Find the acceleration just after the opening, solve for $v(t)$ in stage 2, and compute the time and the distance needed to come down to $1.05\,v_{t2}$. Why can the formula of (b) not be used?

Step 1. Acceleration at the opening

$c_2 = mg/v_{t2}^2 = 9810/64 = 153.3$ kg m^{-1} ($c_2/c_1 = 39.06$). Just after the opening $v = 40$ m s^{-1}, so $\dot v = g\big(1 - (40/8)^2\big) = -24g = -235.4$ m s^{-2}: the drag $c_2 v_1^2 = 245.3$ kN is 25 times the weight.

Step 2. Why not tanh

The formula $v = v_t \tanh(\dots)$ is bounded by v_t, so it can never start at $v_1 = 5v_{t2}$; equivalently the constant of integration would be artanh 5, which is not real. For $v > v_t$ the antiderivative of $1/(1 - v^2/v_t^2)$ is $v_t\,\operatorname{arcoth}(v/v_t)$: $\dfrac{d}{dv}\big[v_t\,\operatorname{arcoth}(v/v_t)\big] = \dfrac{1}{1 - v^2/v_t^2}$, the same derivative as artanh on the other branch.

Step 3. Integration

From $(t = 0, v = v_1)$ to (t, v):

$$\Big[v_{t2}\,\operatorname{arcoth}\frac{v'}{v_{t2}}\Big]_{v_1}^{v} = \big[g\,t'\big]_0^t \;\Longrightarrow\; v = v_{t2}\coth\Big(\frac{gt}{v_{t2}} + \operatorname{arcoth}\frac{v_1}{v_{t2}}\Big)$$

$\coth > 1$ and decreases, so v decreases monotonically to v_{t2} from above.

Step 4. Time to $1.05\,v_{t2}$

$t = \dfrac{v_{t2}}{g}\Big(\operatorname{arcoth} 1.05 - \operatorname{arcoth} 5\Big)$, with $\operatorname{arcoth} 1.05 = \tfrac12 \ln 41 = 1.8568$ and $\operatorname{arcoth} 5 = \tfrac12 \ln 1.5 = 0.2027$: $t = 0.8155 \cdot 1.6541 = 1.349$ s.

Step 5. Distance

As in (c), $\dfrac{v\,dv}{1 - v^2/v_{t2}^2} = g\,dy$, now integrated from $(\Delta y = 0,\ v = v_1)$ to $(\Delta y,\ v)$. Substitute $w = 1 - v'^2/v_{t2}^2$, $dw = -\dfrac{2v'}{v_{t2}^2}dv'$, so $v'\,dv' = -\dfrac{v_{t2}^2}{2}dw$; the limits become $w_1 = 1 - v_1^2/v_{t2}^2 = -24$ and $w = 1 - v^2/v_{t2}^2$. In stage 2 $v' > v_{t2}$, so $w < 0$ on the whole interval and the antiderivative of $1/w$ is $\ln|w|$ (not $\ln w$):

$$g\,\Delta y = -\frac{v_{t2}^2}{2}\int_{w_1}^{w}\frac{dw'}{w'} = -\frac{v_{t2}^2}{2}\Big[\ln|w'|\Big]_{w_1}^{w} = -\frac{v_{t2}^2}{2}\big(\ln|w| - \ln|w_1|\big) = \frac{v_{t2}^2}{2}\ln\frac{|w_1|}{|w|}$$

With $|w_1| = (v_1^2 - v_{t2}^2)/v_{t2}^2$ and $|w| = (v^2 - v_{t2}^2)/v_{t2}^2$ the factors v_{t2}^2 cancel in the quotient; solving for Δy and, inversely, for v^2:

$$v^2 = v_{t2}^2 + \left(v_1^2 - v_{t2}^2\right) e^{-2g\Delta y / v_{t2}^2}, \qquad \Delta y = \frac{v_{t2}^2}{2g} \ln \frac{v_1^2 - v_{t2}^2}{v^2 - v_{t2}^2}$$

With $v = 8.4$ m s^{-1}: $\Delta y = 3.262 \cdot \ln(1536/6.56) = 3.262 \cdot 5.456 = 17.80$ m.

KEY RESULTS

- $c_2 = 153.3$ kg m^{-1}; $\dot v(0^+) = g\left(1 - v_1^2/v_{t2}^2\right) = -24\,g = -235.4$ m s^{-2} (a **deceleration**; drag 245.3 kN against a weight of 9.81 kN).
- $v = v_{t2} \coth\left(\dfrac{gt}{v_{t2}} + \operatorname{arcoth}\dfrac{v_1}{v_{t2}}\right)$, decreasing towards v_{t2} from above.
- To 8.4 m s^{-1}: $t = 1.349$ s, $\Delta y = 17.80$ m.

Part (f) · 2 points

Work of the drag and energy balance

Is the mechanical energy conserved? Compute the work done by the drag (i) in stage 1, from the release to the opening, and (ii) in stage 2, from the opening to $v = 8.4$ m s^{-1}, from the energy balance; then check (ii) by integrating the drag force along the path.

Step 1. Energy is not conserved

The power of the drag is $(-cv^2\,\mathbf{e}_y)\cdot(v\,\mathbf{e}_y) = -cv^3 < 0$ (Notes §4.4): it removes mechanical energy as long as the capsule moves. Only the weight is conservative ($V = -mgy$ with y downward), so $\dfrac{d}{dt}\left(\tfrac12 mv^2 - mgy\right) = -cv^3$. At the opening the dissipated power is $c_2 v_1^3 = 153.3 \cdot 64\,000$ W $= 9.81$ MW.

Step 2. Stage 1

From $v = 0, y = 0$ to $v_1 = 40$ m s^{-1}, $y_1 = 130.2$ m: $\Delta E_{kin} = \tfrac12 \cdot 1000 \cdot 1600 = 800.0$ kJ and $W_{weight} = mgy_1 = 1277.1$ kJ, so $W_{drag,1} = 800.0 - 1277.1 = -477.1$ kJ.

Step 3. Stage 2 from the energy balance

$\Delta E_{kin} = \tfrac12 \cdot 1000\,(70.56 - 1600) = -764.7$ kJ and $W_{weight} = 9810 \cdot 17.80 = 174.6$ kJ, so $W_{drag,2} = -764.7 - 174.6 = -939.3$ kJ: in under 18 m the main parachute absorbs almost twice what the drogue absorbed in 130 m.

Step 4. Direct integral

$W_{drag,2} = -\int_0^{\Delta y} c_2 v^2\, ds$ with $v^2 = v_{t2}^2 + (v_1^2 - v_{t2}^2)e^{-2gs/v_{t2}^2}$ and $c_2 v_{t2}^2 = mg$:

$$W_{drag,2} = -mg\,\Delta y - c_2\left(v_1^2 - v_{t2}^2\right)\Big[-\frac{v_{t2}^2}{2g} e^{-2gs/v_{t2}^2}\Big]_0^{\Delta y} = -mg\,\Delta y - \frac{m}{2}\left(v_1^2 - v_{t2}^2\right)\left(1 - e^{-2g\Delta y/v_{t2}^2}\right)$$

and by (e), $\left(v_1^2 - v_{t2}^2\right)\left(1 - e^{-2g\Delta y / v_{t2}^2}\right) = v_1^2 - v^2$, so $W_{drag,2} = -\left(mg\Delta y + \tfrac{1}{2}m(v_1^2 - v^2)\right) = -939.3$ kJ, the value of Step 3.

KEY RESULTS

- No: the drag power $-cv^3 < 0$ dissipates energy (9.81 MW just after the opening).
- (i) $W_{drag,1} = \tfrac{1}{2}mv_1^2 - mgy_1 = 800.0 - 1277.1 = -477.1$ kJ.
- (ii) $W_{drag,2} = \tfrac{1}{2}m(8.4^2 - 40^2) - mg\Delta y = -764.7 - 174.6 = -939.3$ kJ; the direct integral gives $-\left(mg\Delta y + \tfrac{1}{2}m(v_1^2 - v^2)\right)$, the same.

Interpretation and checks

- **Dimensional check:** $[c] = \text{N}/(\text{m s}^{-1})^2 = \text{kg m}^{-1}$, so $\sqrt{mg/c}$ is m s^{-1}; gt/v_t and gy/v_t^2 are dimensionless (arguments of tanh and exp); $[cv^3] = \text{W}$.
- **Limits:** for $gt \ll v_t$, $\tanh x \approx x$ gives $v \approx gt$ and $y \approx \tfrac{1}{2}gt^2$ (free fall); for $t \to \infty$ both tanh and coth tend to 1 and $v \to v_t$, from below in stage 1 and from above in stage 2; $y \approx v_t t - \frac{v_t^2}{g}\ln 2$ for large t in stage 1.
- **Common mistakes in the exam:** (1) writing the drag as $+cv^2$ or $-cv$ (wrong sign or wrong law); (2) using $v = v_t \tanh(\dots)$ after the main parachute opens, which would need artanh of a number larger than 1; (3) skipping the chain rule $\dot v = v\,dv/dy$; (4) assuming the terminal speed is reached in a finite time; (5) writing energy conservation with drag acting.